Syllabus for MATH 6260-001 Spring 2026

Representation Theory

Course Description:

Instructor: Dragan Miličić
Office: JWB 333
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Days & Time: MWF 12:55 pm – 1:45 pm
Office Hours: After the class or by appointment.
Textbook: I'll distribute my own notes.

Goals and Objectives: The goal of the course is to explain how one extends basic results of representation theory of finite groups to the setting of compact groups. We shall discuss harmonic analysis on these groups as an extension of classical Fourier analysis. Then we are going to discuss the representation theory of connected compact Lie grops. Finally we shall discuss representation theory of noncompact semisimple Lie groups and give an introduction into current research based on some simple examples.

Course content:

- Review of representation theory of finite groups
- Haar measure on compact groups
- Finite-dimensional representations, their characters and Schur's orthogonality relations
- Peter-Weyl theorem
- Induced representations, Frobenius reciprocity
- Weyl character formula for compact Lie groups
- Borel-Weil theorem for compact Lie groups
- Examples
- Overview of representation theory of semisimple Lie groups